



Preparation and evaluation of fouling-release properties of amphiphilic perfluoropolyether-zwitterion cross-linked polymer films

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ABSTRACT

The biofouling of marine structures presents a problem for maritime industries, including increasing fuel and operational costs. There has been much work developing and evaluating chemically ‘ambiguous’ amphiphilic coatings based upon hydrophobic fluoropolymer and hydrophilic poly(ethylene glycol) (PEG). Many of these coatings have shown good fouling-release performance against diatoms, which present a fouling challenge to commercially-available state-of-the-art silicone-based fouling-release coatings. However, PEG is prone to oxidation, which limits its practical use in the marine environment, and thus an alternative hydrophilic material would be desirable in the future development of amphiphilic coatings. In this regard, zwitterionic materials are emerging as a promising class of hydrophilic antifouling material, which we hypothesized would be a suitable alternative to PEG within amphiphilic coatings. To test this hypothesis, cross-linked amphiphilic films consisting of commercially available perfluoropolyethers and the zwitterionic monomer sulfobetaine methacrylate were developed and studied. The difficulty in formulating these chemically incompatible species was overcome by careful choice of solvents and a series of prototype cross-linked films were prepared in which increasing quantities of zwitterion were systematically incorporated. The fouling-release performance of these films was tested against the diatom *Navicula incerta*, a common microfouling organism responsible for the formation of so-called ‘slime’ layers, and antifouling performance with cypris larvae of two barnacle species, *Balanus amphitrite*, and *Balanus improvisus*. To expand the scope of the study, the clinically-relevant biofilm-forming pathogen *Staphylococcus aureus*, which is a major culprit in the infection and failure of millions of indwelling medical devices, was also evaluated. Results indicate that the incorporation of zwitterion into perfluoropolyethers leads to significant improvements in fouling-release performance and these amphiphilic coatings display potential in fouling-release applications.

1. Introduction

The unwanted fouling of surfaces by marine organisms presents a substantial and ongoing problem, leading to reduced vessel performance and increased costs for various maritime industries [1–3]. For example, the increased roughness on ships’ hulls caused by fouling can increase fuel consumption by 40 % [4]. Traditional solutions have involved antifouling coatings which leach biocides (often based upon tin

or copper) but because of the adverse impact caused by the continuous release of toxic compounds into the marine environment these coatings are either banned or under increasing regulatory scrutiny [5,6].

One alternative is so-called ‘fouling-release’ coatings [7,8]. These typically present low surface energy, smooth surfaces that are designed to minimize interactions with biomolecules by eliminating the ability for strong polar interactions such as hydrogen or ionic bonding. Through dispersive interactions alone, biomolecules can adhere only

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Table 1

Quantities, number of moles and mole ratio of zwitterionic monomer (1) and PFPE (2) employed in the formulations of polymer films.

Sample	Mass of 1 / g	Mass of 2 / g	Mass of 1 + 2 / g	Number of moles of 1 / mmol	Concentration of 1 / mM	Number of moles of 2 / mmol	Concentration of 2 / mM	Monomer 1: Monomer 2 (Stoichiometry)
F1Z0	–	5.40	5.40	–	–	2.87	228	0:1
F1Z1	0.70	4.70	5.40	2.51	200	2.50	198	1:1
F1Z2	1.24	4.16	5.40	4.44	352	2.21	176	2:1
F1Z4	2.01	3.39	5.40	7.20	571	1.80	143	4:1
F1Z1r	0.62	4.16	4.78	2.22	176	2.21	176	1:1
F1Z4r	2.47	4.16	6.63	8.84	702	2.21	176	4:1

very weakly to these surfaces, making their removal easier, ideally by ‘self-cleaning’ via the hydrodynamic forces resulting from vessel movement. Coatings based upon silicone elastomers have been particularly well-studied, and there are now numerous commercially successful silicone-based coatings on the market [9,10]. Although generally very effective, especially with hard fouling, such coatings can accumulate slime dominated by microalgae and bacteria [11]—which tend to adhere to hydrophobic surfaces and are not easily released by hydrodynamic shear [12]. This adhered slime generates hydrodynamic drag on hulls moving through the water [13], adding to fuel costs.

To address this issue with silicone-based fouling-release coatings, there has been research interest in chemically ‘ambiguous’ surfaces based upon both hydrophilic and hydrophobic materials, typically fluoropolymers covalently cross-linked with PEG-based polymers [14–26]. Fluoropolymer-based materials present low surface energies, hydrophobicity, oleophobicity, and possess excellent thermal and chemical stability. PEGylated materials, on the other hand, present long flexible hydrophilic chains, which being well hydrated, are very effective at resisting protein adsorption [27–31]. When presented together, the resulting amphiphilic coatings are able to benefit from the virtues of both types of polymer and good resistance to fouling organisms such as *Navicula incerta* and *Ulva* spp. has been reported [16–26]. PEG-based materials, however, are readily subject to oxidation, especially in the presence of oxygen and transition-metal ions, which limits the practical utilization of this class of polymer [32]. Alternatives to PEG include hydrophilic materials based upon zwitterions, chemical moieties containing both positive and negative charged units. Similarly to PEG-based surfaces, zwitterions possess the ability to strongly bind water molecules via electrostatic interactions, forming a hydration layer. This removes any thermodynamic advantage from the adsorption of biomolecules, because it is energetically favourable for the surface to remain in contact with water rather than an amphiphilic biomolecule e.g. a protein secreted by an organism [33–36]. Consequently, zwitterionic polymers are also highly resistant to protein adsorption and cell adhesion [33,37–39].

In this work we set out to prepare and evaluate the antifouling and fouling-release efficacy of prototype amphiphilic polymer films composed of fluoropolymer and zwitterions. We chose to utilize perfluoropolyether (PFPE) as fluoropolymer components, which make attractive hydrophobic building blocks for functional coatings on account of their low surface energies, very good chemical resistance and high thermal stability. Furthermore, they are biodegradable and do not release environmentally persistent degradation products [40,41], unlike polymers based upon pendant perfluorooctanoic acid, which can bioaccumulate in tissues [42]. As the zwitterionic component, we chose a well-studied and commercially-available sulfobetaine monomer. We overcame the issue of formulating two chemically very incompatible species and explored a selection of different PFPE-zwitterion compositions, characterizing the films by a range of techniques. We investigated their antifouling and fouling-release properties with model marine organisms (*Balanus amphitrite*, *Balanus improvises* and *Navicula incerta*) and also *Staphylococcus aureus*, a common bacterium which is a well-known source of fouling of biomedical implants. Results indicate that ‘amphiphilic’ PFPE-zwitterion-based materials may be effective as

antifouling and fouling-release coatings and are worthy of further investigation.

2. Experimental section

2.1. Materials and chemicals

The zwitterionic monomer [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (1) was obtained from Sigma-Aldrich Company (Gillingham, Dorset, UK) and used as supplied. The commercially available α , ω -bis-methacrylate perfluoropolyether (M.Wt. 1.88 kg/mol) (2) (marketed as Fluorolink MD700®) was provided by Solvay SpA, Bollate, Italy. Trifluoroethanol was purchased from Apollo Scientific. Artificial seawater (ASW) at 32 practical salinity unit (PSU) was prepared, as per manufacturer instructions, from Tropic Marin® Sea Water Classic. All other solvents and chemicals were obtained from Sigma-Aldrich Company, unless indicated otherwise, and used as supplied.

2.2. Film preparation

Copolymer films were prepared in mixtures of trifluoroethanol and water via the radical copolymerization of the zwitterionic monomer, [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (1) with the diacrylate PFPE macromonomer (2) in the mole ratios 0:1, 1:1, 2:1 and 4:1 (Table 1). Each component was dissolved in 5.4 mL trifluoroethanol, the solutions were combined, 1.8 mL of distilled water was added, and the mixture stirred to obtain a homogeneous solution. The proportions of trifluoroethanol and water were found to be important in order to maintain the solubilities of all components. A freshly prepared ammonium persulfate (APS) solution in water (160 μ L, 4.4 M) was added immediately followed by tetramethylethylenediamine, (TMEDA) (0.36 mmoles). The formulations were homogenized by agitation with a stir bar and then injected into a sandwich of two glass Petri dishes separated by 1.0 mm using glass microscope slides as spacers (Fig. S1), which presents a straightforward method to prepare materials with flat surfaces that are a prerequisite for biological settlement assays. The formulations were allowed to react for 12 h at room temperature and the resulting films were then soaked in MeOH for 1 h followed by water for 1 h. The samples were removed from water and the water soaking procedure was repeated a further two times in fresh water. During these steps, the samples shrink slightly which likely causes an increase in their surface roughness. The samples were stored in distilled water and removed when required for analysis. The samples were stored in distilled water for two months and no change in their properties was observed during this time, suggesting the samples are very stable under water. However, when removed from the water they dried (over about 48 h), became brittle and cracked. We use the notation FxZy to describe the samples, where x and y are the stoichiometric ratios of PFPE (2) to zwitterionic monomer (1). In order to determine a yield (the percentage of 1 and 2 which are converted into polymer film), samples were left to dry in a desiccator, then weighed. The yield was calculated as:

(mass of polymer film/total mass of $1 + 2$) $\times 100$ % and was determined to be 84 ± 3 % in all cases.

2.3. Water contact angle

The static contact angle of the wet samples under distilled water was determined by the captive bubble method. The angle between liquid/vapour and solid/ liquid interface of the samples was measured using a KSV CAM 101 goniometer (KSV Instruments Ltd., Helsinki, Finland) at room temperature. The samples were placed upside down in the glass chamber and an air bubble with a volume of $\sim 2 \mu\text{L}$ was placed on the film surface (Fig. S2a). Contact angle measurements are given as a mean of two repeats recorded in each of 10 different positions, with variance expressed as $\pm$ one standard deviation from the mean. Since samples were stored in water there was no need to pre-immersed the sample prior to contact angle measurement.

2.4. Surface analysis by atomic force microscopy (AFM)

AFM Images and roughness data were acquired using a Dimension Model D3100 V (Veeco) atomic force microscope with a Nanoscope V controller (Bruker) and a large scanner ($\sim 100 \mu\text{m}^2$). Nanoscope software v720 was used to control the microscope. The system was operated in tapping mode under water. For reducing vibrational noise, an isolation table/acoustic enclosure was used (Veeco Inc., Metrology Group). Silicon tips on silicon cantilevers (MPP-21100-10) were used for imaging. The nominal tip radius was $\sim 8 \text{ nm}$, resonant frequency 75 kHz , and spring constant $k \ 3 \text{ N/m}$. The AFM data were analysed with NanoScope Analysis 1.5 software (Bruker). Roughness was calculated on a sample area of $100 \mu\text{m}^2$. The root mean square of roughness (R_q) was calculated by taking the average of measurements obtained from 20 randomly selected areas of each coating under water.

2.5. Nanomechanical properties

Nanomechanical data were calculated from force curves that were acquired using a Digital Instruments Multimode-8 with Nanoscope V controller and E scanners (Bruker, Germany). Nanoscope software version 9.1 was used to control the microscope. NanoScope Analysis 1.50 software (Bruker) was used to process the force curves. Silicon tips on V-shaped Si_3N_4 cantilevers (model DNP, Bruker) were used for the force measurement. The AFM probes were cleaned by exposure to oxygen plasma for 10 min at 70 W. The force curves were obtained by operating the AFM in Force Volume-Contact mode. The force curves were used to determine adhesion strength, Young's modulus and stiffness. The deflection sensitivity of each cantilever was calibrated by performing closed-loop Z force curves on flat silicon. A RS-12 M titanium sample (Bruker, Germany), with very sharp grain features, was used to estimate the radius of curvature of the AFM probes. The spring constant of each cantilever was calibrated using the thermal tune method [43]. All experiments were carried out in an AFM fluid cell and under water. Young's modulus of elasticity of the samples was estimated by processing the force curves that were acquired by AFM on the coatings' surfaces under water. The force curves were obtained at 200 locations on each sample with the AFM operated in Force Volume-Contact mode under water.

2.6. X-ray photoelectron spectroscopy

X-Ray Photoelectron Spectroscopy (XPS) was performed at the XPS Users Service (NEXUS) at Newcastle University. A Thermo Scientific K-Alpha X-ray photoelectron spectrometer (Thermo Electron Corp., East Grinstead, UK) was used to acquire XPS spectra. Samples were removed from their storage medium (distilled water), excess water was removed by gently patting with tissue paper and the samples were then left to

dry for 24 h in air. XPS survey and high-resolution scan spectra were analysed using the CasaXPS software (Casa, <http://www.casaxps.com>, UK). All spectra were corrected to hydrocarbon C1s peak at 285 eV as reference. Wide scan spectra were recorded at a pass energy of 150 and 1 eV / step, while narrow scan spectra were recorded at a pass energy of 50 and 0.1 eV/step. Depth profile XPS spectra were acquired after 300 s of sputtering using argon gas cluster ion.

2.7. FTIR and CHN analyses

FTIR Spectra (in the range of 400 to 4000 cm^{-1} wavenumbers) were recorded using the ATR accessory (diamond crystal) of an IRAffinity-1S Fourier transform infrared spectrophotometer at 4 cm^{-1} spectral resolution. For each spectrum, 64 scans were co-added and averaged. The sample was measured as a dry polymer film and the bare ATR accessory was used as a background. Spectral references (sulfobetaine) were obtained from the AIST Spectral Database for Organic Compounds (http://sdb.sdb.aist.go.jp/sdb/cgi-bin/cre_index.cgi). The service provided by the School of Human Sciences, London Metropolitan University was used to perform elemental analysis.

2.8. Rheological measurements

Rheological measurements were performed with a HR-2 Discovery Hybrid Rheometer (TA Instruments) with a standard steel parallel-plate geometry of 20 mm diameter with a gap of 0.6 mm. The strain and the frequency were set to 1 % and 1 Hz, respectively.

3. Biological evaluation

3.1. Bacterial biofilm assay

Staphylococcus aureus NCTC6571 was routinely cultured in Tryptone Soy Broth (TSB; Melford Biolaboratories Ltd, Ipswich, UK) at 37 °C with shaking at 200 rpm. Stock cultures were stored at -80 °C in 50 % glycerol / 50 % TSB. For biofilm growth experiments, 10 μL of stock culture were added to 2 mL of TSB in one well of a plastic 24 well plate (Greiner Bio-One, Frickenhausen, Germany) containing a circular piece of polymer film sample (14 mm diameter). The samples were incubated statically at 37 °C for 24 h in a humid environment. After incubation, the supernatant was removed and each sample of film was washed three times with 1.0 mL of phosphate buffered saline (PBS, pH 7.4, Severn Biotech Ltd, Kidderminster, UK). A further 1.0 mL of PBS was added and the cells were gently scraped from the upper face of the polymeric material using a tissue culture scraper. Samples were serially diluted 10-fold in PBS to a 10^{-6} dilution, and 60 μL of each dilution was spread over a TSB agar plate. After incubation at 37 °C for 24 h, the colony forming units (CFUs) were calculated per mm^2 of polymer film by counting the number of colonies on an appropriate dilution. Biofilm assays were performed four times on different days, each time in duplicate. Samples for scanning electron microscopy (SEM) were incubated in a similar way, except the polymer films were not scraped; instead they were fixed in 2 % glutaraldehyde in Sorensen's Phosphate Buffer, dehydrated through a series of graded ethanol before being critical-point dried with CO_2 (Baltec) and coated with 15 nm of Au using a Polaron SEM coating unit. Images were taken using a Tescan Vega SEM operated at 8 kV.

3.2. Diatom initial adhesion and ease-of-removal assay

Adhesion and ease-of-removal assays were carried out using the diatom *Navicula incerta*. Diatoms were cultured in F/2 medium in 250 mL conical flasks and harvested while in log phase growth (after 3–4 d). In preparation for assays, diatom cells were resuspended in 0.22 μm -filtered ASW and diluted to an optical density of 0.02 at 660 nm. Prior to testing, all standard materials included for comparison were

wetted in deionised water for 24 h and then transferred to artificial seawater (ASW) for a further 24 h. Polymer film samples were stored fully hydrated in ASW.

An automated, calibrated, water jet system [44] was used to test diatom cell adhesion on all six polymer surfaces. Six pieces of each polymer film were cut to approximately the same dimensions as a standard microscope slide (76×26 mm) and placed in quadriPERM (Sarstedt, Nümbrecht, Germany) dishes, along with six glass slides coated with polydimethyl siloxane (PDMS; Dow Corning 3-0213, Dow Corning Corporation, Auburn MI, USA) and six uncoated glass slides. 10 ml of diatom suspension was then added to each quadriPERM dish well, and the dishes were left in ambient light conditions at room temperature for 2 h to allow diatom settlement and adhesion. All samples were then gently rinsed to remove unattached cells, by flooding the dishes with clean ASW and placing them on an orbital shaker at 60 rpm for 5 min. After this, three replicates of each surface were exposed to the water jet at an impact pressure of 23 kPa. All replicates were then imaged whilst wet under fluorescence microscopy (Leica DMi8, Leica Microsystems GmbH, Wetzlar, Germany), with illumination at 546 nm (excitation) / 590 nm (emission). The diatom cell density was taken as the mean number of cells counted in 15 fields of view, divided by the calculated area of the field of view (0.6 mm^2). Percent removal was calculated as the reduction in diatom cell density on water-jet-exposed samples compared to the mean diatom density of the non-exposed surfaces.

3.3. Barnacle settlement assays

Adult barnacles, *Balanus amphitrite* (= *Amphibalanus amphitrite*), were induced to release nauplii, which were reared to the cyprid stage following established protocols [45,46], except that all culture stages were carried out in 32 PSU artificial seawater, and nauplii were reared on *Tetraselmis suecica*. Post-metamorphosis, cyprids were stored in 0.22 μm -filtered ASW for 3 d at 6°C prior to use in settlement assays.

Two *B. amphitrite* settlement experiments were carried out. The first compared settlement on F1Z1, F1Z2 and F1Z4, while the second compared settlement on F1Z0, F1Z1r, F1Z2 and F1Z4r. For both experiments, 20 mm diameter discs of the polymer films were cut and placed flat on a hydrophobic backing material (Intersleek® 757, provided by AkzoNobel). Approximately 750 μl of 0.22 μm -filtered ASW was carefully pipetted onto the surface of each disc, adjusting the volume to form a droplet over the whole polymer disc. Twenty (± 2) cypris larvae were added to the droplet in a minimal volume of ASW using a Pasteur pipette. The hydrophobic backing material prevented lateral spread of the droplet, so that cyprids were confined above the polymer film. During both experiments, settlement was simultaneously assessed on PDMS-coated glass slides for comparison, with the 750 μl water droplets placed directly on the PDMS surface. All samples were then incubated at 28°C in conditions of high humidity, in the dark. Numbers of settled cyprids were recorded after 48 h, and the proportion settled was calculated, excluding any cyprids which had escaped over the edges of the polymer film discs. Any replicates where more than 50 % of the cyprids had escaped, or where the droplet water level had fallen such that cyprids were unable to explore the test surfaces, were excluded. The target number of replicates for each surface was 10; final replicate numbers after these exclusions are given in the results.

This assay was repeated on surfaces F1Z0, F1Z1r, F1Z2, and F1Z4r, using a second barnacle species, *Balanus improvisus* (= *Amphibalanus improvisus*). The protocols were identical to those for *B. amphitrite*, except that adult barnacles were maintained at 15°C , all culture stages were carried out using lower salinity ASW (22 PSU), and 30 cyprids were used for the settlement assay (also using 22 PSU ASW), immediately after metamorphosis.

3.4. Biological data analysis

All biofouling assay data analyses were conducted using R version 3.5.1 [47]. Since the bacterial biofilm data were obtained from four separate experimental repeats, the data were combined and analysed via a meta-analysis approach [48] using the *metafor* package [49]. The effect size measure used for the meta-analysis was the response ratio relative to the zwitterion-free control surface (F1Z0); for each zwitterionic surface this was calculated as the natural logarithm of the mean CFU on the experimental surface divided by that on the control (F1Z0). Both the response ratio and its sampling variance were calculated using the 'escalc' function in *metafor*. A multi-level meta-analysis model was used to test for variation among surfaces ('surface' was included as a moderator in the meta-analysis model), specifying experiment-level random effects and using weights derived from an appropriate variance-covariance matrix to account for statistical dependence arising from the comparison of multiple surfaces to a single control in each experiment. Full details of meta-analysis modelling and relevant code are presented in Supplementary material.

Diatom assay data (diatom density, proportional removal) were compared among different surfaces using Welch's ANOVA; this adjusted test does not require the assumption of equal variances.

4. Results and discussion

Zwitterionic-PFPE films were prepared by the copolymerization of sulfobetaine methacrylate **1** with α , ω -bis-methacrylate PFPE **2** (Fig. 1a) using APS/TMEDA as initiators in a trifluoroethanol-water solvent. Developing suitable co-solvent mixtures to perform the copolymerization of these two chemically incompatible species required exploration of solvent mixtures, and thus different ratios of trifluoroethanol and water were assessed until a suitable solvent composition of water and trifluoroethanol was found (see Experimental) in which all reaction components dissolved; the resulting homogenous solutions were injected into a sandwich of two glass Petri dishes (Fig. S1) and allowed to react for 12 h at room temperature to afford the desired polymer films. MeOH/water and acetone/water were also investigated as solvent systems but did not provide successful homogeneous formulations over a range of different solvent compositions.

By systematically altering the mole ratios of zwitterion **1** to PFPE **2**, six different zwitterion-fluoropolymer films were obtained as opaque, white, free-standing films containing differing zwitterion contents (Table 1) and whose thicknesses were measured to be in the range 0.5–0.7 mm. We use the notation FxZy to describe the samples, where the PFPE monomer is designated F and sulfobetaine monomer as Z, with x and y designating relative molar ratios. Oscillatory rheology frequency sweep experiments (Fig. S3) show that the storage moduli were always larger than the loss moduli at all frequencies, an observation suggesting that the PFPE chains are acting as chemical cross-linkers within the materials [50]. Experiments indicated that for samples F1Z1, F1Z2 and F1Z4, the sample roughness (Table 2) and sample's Young's modulus (Fig. 3) increased as the amount of **1** was systematically increased. It was unclear if these changes in roughness and Young's modulus were due to the increase in the concentration of **1** or were a consequence of the decrease in the concentration of **2**. We discovered that by changing the concentrations of the monomers (whilst keeping their stoichiometry fixed), it was possible to create additional samples of identical composition but differing roughness and nanomechanical properties (F1Z1r and F1Z4r). We speculate that differences in surface roughness arise mainly on how the cross-linked polymer network responds as the solvent is changed to water from trifluoroethanol/water. Copolymer film samples prepared with higher concentrations of PFPE monomer in the formulation presumably afford denser cross-linked networks. When the samples are placed under water, which is an unfavourable solvent for PFPE, denser cross-linked networks have less scope for network collapse accompanied by concomitant crumpling of their surfaces, and hence

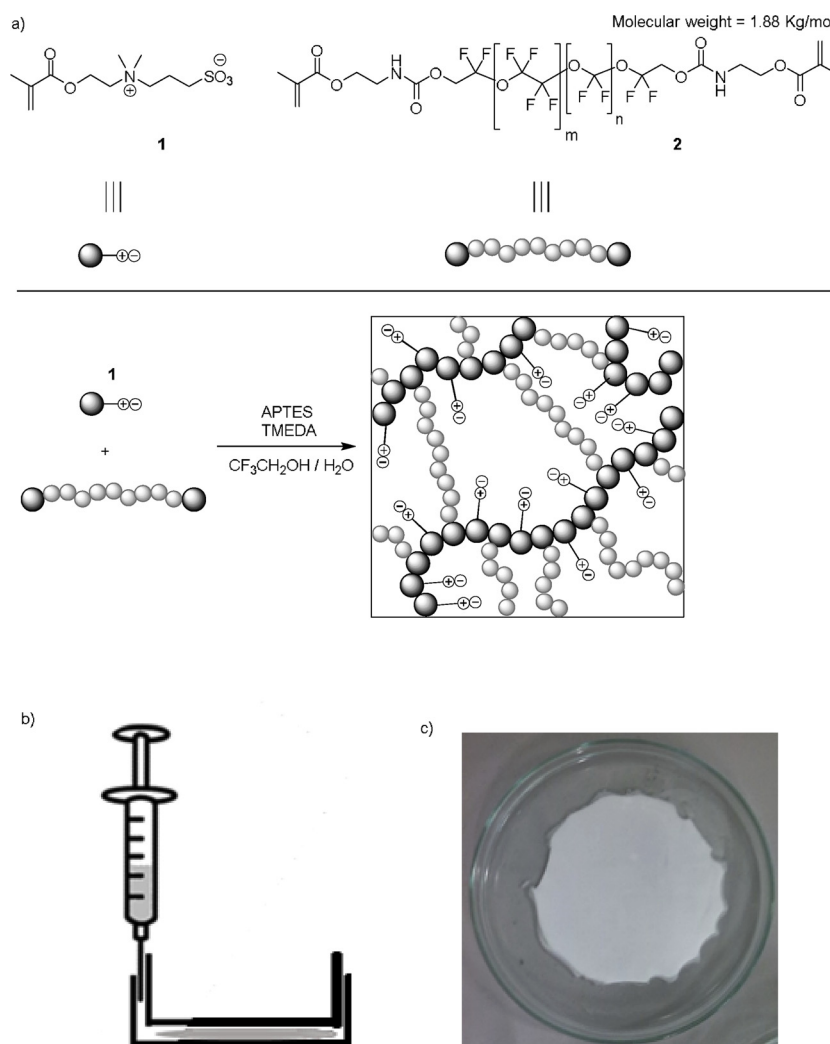


Fig. 1. a) Structures of sulfobetaine methacrylate (**1**) and α,ω -bis-methacrylate PFPE (**2**) (molecular weight 1.88 Kg/mol). b) Copolymerization of **1** and **2** to afford zwitterion-PFPE cross-linked materials. c) The preparation of the cross-linked coatings involves the injection of the monomer formulation between two glass Petri dishes where one dish sits inside another, separated by 1.0 mm. d) Photograph of a representative cross-linked film (**F1Z2**).

Table 2

Summary of molar compositions, roughness and water contact angle for all films. It is important to emphasize that samples **F1Z1** and **F1Z4** have the same composition of **F1Z1r** and **F1Z4r**, respectively.

Film	Content of 1 (%) ^(a)	Content of 2 (%) ^(a)	R _q (μm)	R _{max} (μm)	Air in water contact angle (°)
F1Z0	0	1	1.01 ± 0.04	5.89 ± 0.12	53.0 ± 0.8
F1Z1	1	1	0.28 ± 0.01	1.51 ± 0.07	14.6 ± 1.0
F1Z2	2	1	0.31 ± 0.03	1.69 ± 0.16	12.0 ± 1.1
F1Z4	4	1	0.54 ± 0.01	2.73 ± 0.08	11.9 ± 0.9
F1Z1r	1	1	0.58 ± 0.02	3.00 ± 0.09	16.8 ± 1.0
F1Z4r	4	1	0.46 ± 0.01	2.23 ± 0.04	11.6 ± 0.8

^aPercentage composition on a molar basis. ^b R_q is the root mean square of roughness and R_{max} is the maximum roughness depth.

lower surface roughness (roughness of **F1Z1** < **F1Z2** < **F1Z4**). For comparison of samples prepared with the same stoichiometries, again higher concentration of PFPE in the formulation provides denser cross-linked networks and so lower surface roughness (roughness of **F1Z1** < **F1Z1r**, **F1Z4** > **F1Z4r**).

Complete and unambiguous interpretation of the FTIR spectra (Fig. S4) was not possible because the bands of the different functional groups overlapped. It was possible, however, to observe the presence of

bands which arise on account of the stretching of the bonds of S=O at 1034 and 1190 cm⁻¹ for those samples containing zwitterion, but no quantitative conclusions can be made. However, CHN elemental analysis (Table S1) of each sample revealed an excellent match between the theoretical and actual elemental compositions, indicating that the monomer composition of each film matched closely the monomer feed ratios.

Air-in-water contact angle measurements (Table 2, Fig. S2b) suggest that, as expected, the hydrophilicity of the coatings increased when the zwitterionic monomer was incorporated into the cross-linked polymer film. The measured contact angles of the films incorporating zwitterion **1** display relatively little variation (Fig. S2b); this suggests all of these samples display a relatively constant level of zwitterions at their surfaces even though their monomer compositions vary.

AFM topographical imaging of all films was performed on a sample area of 100 μm² to provide information on their surface topologies and roughness. The AFM image of **F1Z0** (Fig. 2), which is composed of only PFPE **2**, displayed a very rough surface with R_q = 1097 ± 35 nm. Samples that contain both zwitterion **1** and PFPE **2** displayed reduced values of roughness (R_q = 275–575 nm) and different surface topologies (Fig. 2, Fig. S5), suggesting that the incorporation of **1** into the samples altered the nature of the surface of the cross-linked films. The reason for this difference in roughness is unclear, however, one possibility is that in the preparation of **F1Z0** rougher surfaces are formed to

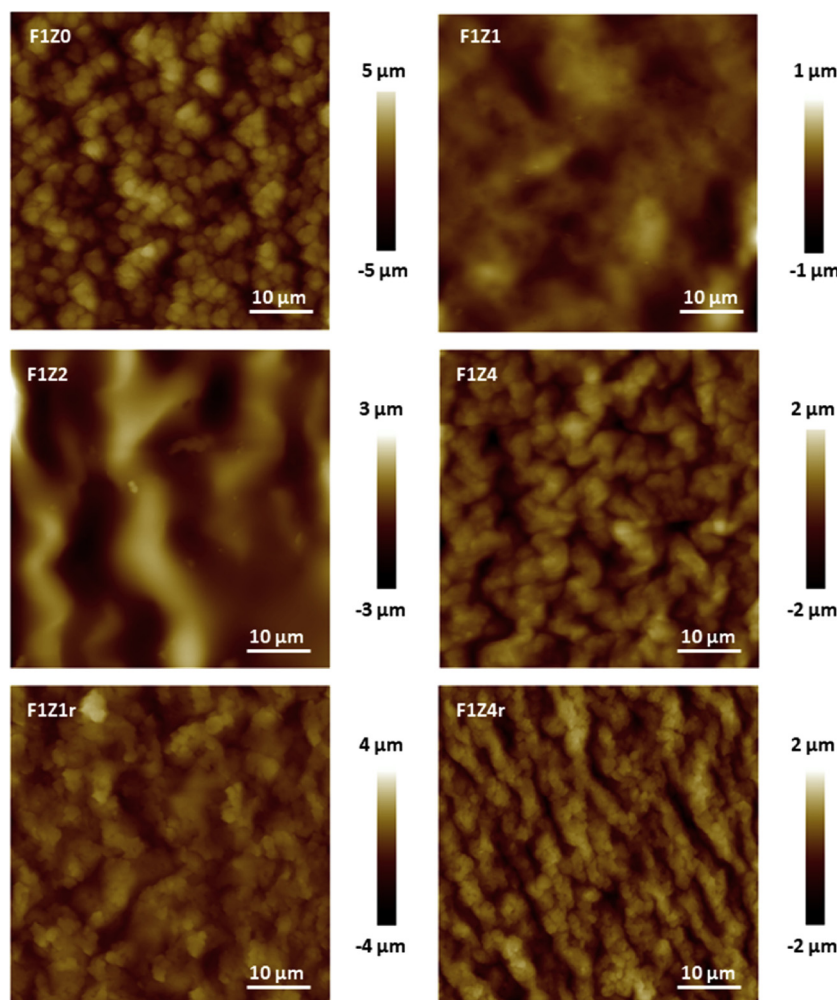


Fig. 2. Representative AFM topographical images of F1Z0, F1Z1, F1Z2, F1Z4, F1Z1r and F1Z4r films. Fig. S5b shows these same images but with an identical scale bar ($\pm 5 \mu\text{m}$).

minimize surface contact between the hydrophobic PFPE with the hydrophilic glass surface.

Values of Young's modulus were also obtained by AFM analysis to provide insight into the mechanical nature of the cross-linked polymer films. The Young's modulus (Fig. 3) was reduced when zwitterionic **1** was incorporated, suggesting incorporation of **1** decreases the stiffness of the cross-linked films; this observation was anticipated because the relative amount of cross-linking macromonomer within the film decreases as the amount of zwitterion increases. Interestingly, films exhibiting identical compositions (F1Z1 vs F1Z1r and F1Z4 vs F1Z4r) display different values of Young's modulus, showing that the method of sample preparation also influences Young's modulus and that this does not correlate with the amount of **1**. To gain further insight as to the macroscopic mechanical properties, rheological measurements were performed (Fig. S3, Table S2).

XPS of all samples was performed to determine the chemical composition of the coating samples at the surface and into the bulk of the films (Figs. S6–9). Survey scan spectra (Fig. S6 and Table S3) revealed that the composition of C, N, O and F was relatively invariant, and the measured percentages of C and N did not match those determined by CHN analysis. It is possible that under the conditions of XPS analysis (when the surface is dried out) the surface is dominated by PFPE **2** with little zwitterion displayed. XPS analysis did indicate a higher F content at the surface than at depth (Table S3). Furthermore, sulfur was not detected at the surface of any of the samples, and only detected by XPS at depth in the sample F1Z4r, which contains the highest amount of

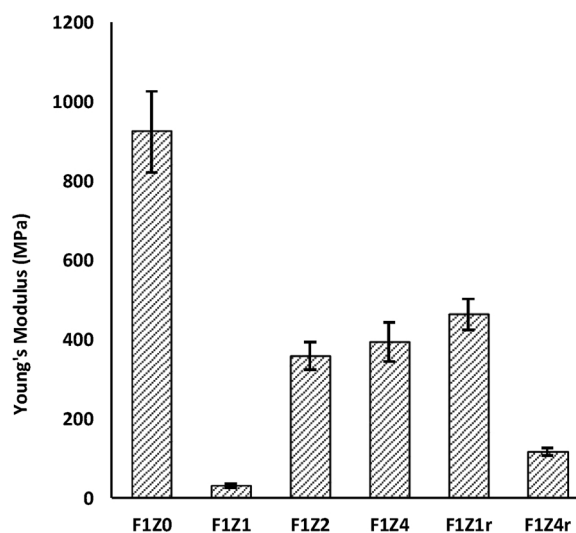


Fig. 3. The Young's Modulus of polymer films under water measured using Si_3N_4 AFM probes.

zwitterion **1**. When the surfaces are under water, it is more likely that they are dominated by zwitterion; this idea is supported by contact angle measurements.

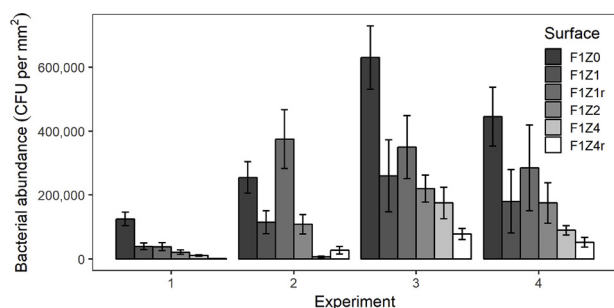


Fig. 4. Bacterial density (colony forming units, CFU, per mm²) measured in each of the four experimental repeats (Experiments 1–4, x-axis). Error bars are $\pm$ standard deviation, $n = 2$ for each surface in each experiment.

4.1. Bacterial biofilm assay

Bacterial density (Fig. 4) generally appeared to be lower on zwitterion-containing films compared to pure PFPE (F1Z0), although in some experiments bacterial abundance on F1Z1r approached or exceeded that on F1Z0. Combining the results of the four experiments via meta-analysis (Table S4) indicated that there was significant variation in bacterial abundance among the different zwitterion-containing formulations (Significance test of moderator ‘surface’, $Q_{Mdf\ 4} = 27.41$, $p < 0.0001$); all surfaces had significantly lower bacterial density than F1Z0 ($p < 0.05$, Table S4) except for F1Z1r ($p = 0.30$). For the films containing zwitterion 1, CFU appeared to decrease as the zwitterion 1 level increased; although there was no significant difference between bacterial density on F1Z1 and F1Z2 (pairwise test, Tukey method, $p = 0.67$), the lowest CFU measurements were always from the surfaces with the greatest zwitterion content: F1Z4 and F1Z4r. The meta-analysis model (Table S4) estimated that the number of CFUs on F1Z4 was 11 % [95 % CI 2 %, 25 %] of that on F1Z0, and on F1Z4r it was 6 % [95 % CI 2 %, 13 %] of that on F1Z0 – an approximately 90 % or greater reduction in bacterial density when the zwitterion was present. Considering pairs of samples with the same level of zwitterion content (F1Z1 versus F1Z1r, and F1Z4 versus F1Z4r), in both cases the sample with the lowest roughness appeared to have fewer bacteria (Fig. 4, S10), although these differences were not statistically significant at these sample sizes (pairwise test, Tukey method: F1Z1 versus F1Z1r, $p = 0.42$; F1Z4 versus F1Z4r, $p = 0.30$). The difference in roughness between F1Z1 and F1Z1r was much larger than that between F1Z4 and F1Z4r (Table 2), and the difference in normalised CFU was correspondingly larger between the former pair. Considering the two surfaces with the highest roughness (F1Z4 versus F1Z1r), the surface with the higher zwitterionic content (F1Z4) had a significantly lower CFU (pairwise test, Tukey method, $p = 0.0049$), lower even than smoother surfaces with lower zwitterionic content (F1Z1 and F1Z2).

4.2. Diatom adhesion and ease-of-removal assay

The apparent differences among surfaces in initial density of *N. incerta* prior to the water jet treatment were not statistically significant (ANOVA; $df = 7, 16$; $F = 1.65$, $p = 0.19$; Fig. 5a). For the assay used here, diatoms settle under gravity, and substantial differences in initial adhesion among surfaces were not anticipated; the lack of significant differentiation among surfaces was thus not surprising. All samples were gently rinsed to remove non-adherent cells before microscopic examination, so it is possible for differences between samples to emerge during this process if adhesion to some surfaces is particularly weak. Although all differences were non-significant, there were some potentially suggestive patterns (Fig. 5a), which show some similarities to the results of the bacteria assay described above (Fig. 4). In agreement with the bacterial results, when comparing pairs of surfaces with the same zwitterionic content (F1Z1 versus F1Z1r, and F1Z4 versus F1Z4r) it

appeared that for relatively low zwitterionic content, the density of diatom cells was higher on the rougher surface. However, for the pair of surfaces with the higher zwitterionic content (F1Z4 and F1Z4r) diatom densities were almost identical (the difference in roughness between these two surfaces was comparatively small). When comparing the two zwitterionic surfaces with the highest roughness (F1Z4 and F1Z1r), the density appeared lower on the surface with higher zwitterionic content (F1Z4). Overall, the lowest densities appeared to be observed on the surfaces with the lowest roughness (F1Z1) or highest zwitterion content (F1Z4 and F1Z4r). Using a calibrated water jet system (23 kPa impact pressure), removal of diatoms from zwitterionic surfaces was very high (Fig. 5b); after exposure to the water jet, very few diatoms remained present on these surfaces. It was therefore not possible to distinguish among the zwitterionic surfaces in terms of the final density of diatoms, or the proportion removed by the water jet. Nevertheless, this clearly indicates that the inclusion of the zwitterionic component in the films reduced the adhesion strength of the diatoms, since only around 60 % of diatoms were removed from F1Z0 at the same impact pressure. Similarly, a higher proportion of diatoms were released from the zwitterionic surfaces compared to two standard surfaces: glass (87.8 % removal) and PDMS (69.4 % removal).

4.3. Barnacle settlement assays

Barnacle settlement was extremely low on the polymer films in all experiments (both species), with zero settlement for most surfaces in most tests, and with mean settlement on each surface always being less than 3 % (Table 3). These observations suggest that the experimental surfaces were very effective at minimising the settlement of both barnacle species. Previous studies have recorded no settlement of *B. amphitrite* on zwitterionic surfaces [50] but the lack of settlement on F1Z0 (pure PFPE) indicates that the present result may not simply be an effect of the zwitterion. Previous evidence has also shown that the settlement preferences of the two barnacle species are not identical: Dahlstrom *et al.* found [51] that settlement of *B. improvisus* was higher on more hydrophobic surfaces, while Finlay *et al.* found [52] the opposite pattern for *B. amphitrite*. This suggests that the prevention of barnacle settlement in this study is also not simply caused by the hydrophilic nature of the surfaces. The lack of settlement also prevented any tests of barnacle adhesion strength. However, during the second experiment with *B. amphitrite*, a small number of barnacles were observed that had metamorphosed but were not attached to the surface (3 out of 197 for F1Z2, 3 out of 171 for F1Z1r, and 1 out of 119 for F1Z4r). This may indicate that some barnacles settled on the surface but were so weakly adhered that they were unable to remain attached.

Overall, the results of the laboratory assays demonstrate that these materials show promise as antifouling and fouling-release surfaces. The performance of the surfaces in the bacteria and diatom assays appeared to be influenced both by the zwitterionic nature of the surface and differences in roughness; the best performing surfaces were those with the highest zwitterionic content and the lowest roughness, while the worst performing zwitterionic surface was the one with the highest roughness and lowest zwitterion content (F1Z1r). Consideration of possible correlations of antifouling assay results with physico-chemical properties of surfaces (Figs. S11, S12) suggests that diatom and bacterial abundance were generally lowest (and proportional removal of diatoms was greatest) on surfaces with the lowest roughness, lowest contact angle, and lowest Young’s modulus. This is generally in agreement with existing evidence; for example, adhesion and growth of *S. aureus* [53–55], and *N. incerta* [52] are reduced on more hydrophilic surfaces.

5. Conclusions

In this study we have crosslinked (without the use of UV-mediated radical initiator) two chemically incompatible monomers—sulfobetaine

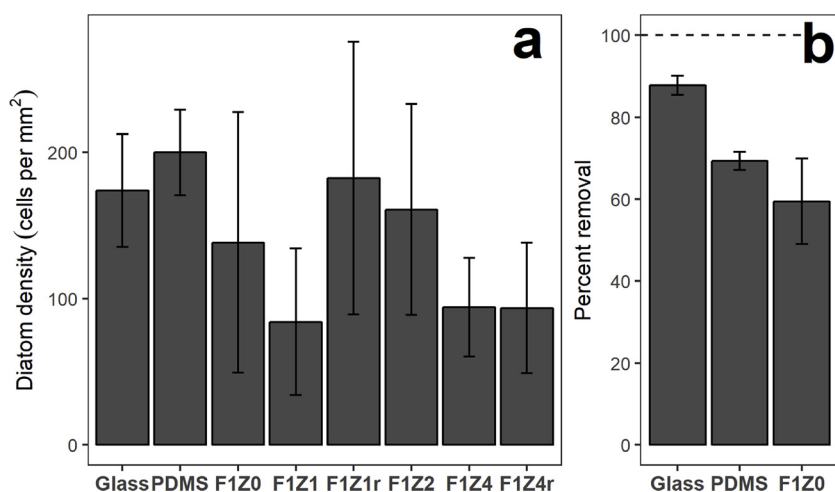


Fig. 5. Diatom adhesion and ease of release data. **a)** mean density of diatoms on all surfaces prior to water jet exposure. **b)** Percent removal from glass, PDMS and **F1Z0** after exposure to automated water jet (23 kPa impact pressure), removal was effectively 100 % from all other test surfaces (indicated by dashed line). All error bars are $\pm$ standard deviation.

Table 3

Barnacle settlement data: proportion of cyprids settled $\pm$ standard deviation. n = the number of replicates of each surface tested. nt = surface not tested during experiment.

Surface	Barnacle settlement		
	<i>B. amphitrite</i> Expt A	<i>B. amphitrite</i> Expt B	<i>B. improvisus</i>
PDMS	0.56 $\pm$ 0.19 (n = 14)	0.62 $\pm$ 0.22 (n = 10)	0.14 $\pm$ 0.15 (n = 10)
F1Z0	nt	0 (n = 2)	0 (n = 10)
F1Z1	0.01 $\pm$ 0.02 (n = 5)	nt	nt
F1Z2	0 (n = 6)	0 (n = 10)	0 (n = 10)
F1Z4	0.02 $\pm$ 0.04 (n = 8)	nt	nt
F1Z1r	nt	0 (n = 9)	0 (n = 10)
F1Z4r	nt	0 (n = 7)	0.01 $\pm$ 0.03 (n = 5)

and PFPE dimethacrylate—to form new cross-linked polymer films. The results of the laboratory biofouling assays demonstrate that these types of polymer films show promise as antifouling surfaces, and further support the idea that by imparting characteristics of both hydrophilicity and hydrophobicity, amphiphilic materials may effectively act as both antifouling and foul release materials. Settlement of two barnacle species with potentially divergent surface preferences was reduced to near-zero. All zwitterion-containing surfaces displayed very low diatom adhesion strength, while settled densities of bacteria were greatly reduced at higher levels of zwitterionic content. While zwitterionic content clearly had an influence on antifouling performance, the tests also indicated a strong effect of surface roughness. We have observed previously [56] that the incorporation of zwitterions into lauryl methacrylate—a hydrocarbon-based hydrophobic monomer absent in fluorine coatings leads to only minor enhancement in antifouling against *N. incerta*, suggesting that the pairing of a hydrophobic fluoropolymer with hydrophilic zwitterion is required to obtain an amphiphilic film with enhanced antibiofouling properties.

Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.porgcoat.2019.105524>.

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